

Math CS 121 HW 8

Zih-Yu Hsieh

December 2, 2025

1

Question 1. 8.3.7:

Let X and Y be continuous random variables with joint probability density function given by

$$f(x, y) = \begin{cases} e^{-x(y+1)} & x \geq 0, 0 \leq y \leq e - 1 \\ 0 & \text{elsewhere} \end{cases}$$

Calculate $\mathbb{E}[X|Y = y]$.

Proof. First, we'll gather the marginal density of Y : If $t < 0$ or $t > e - 1$, since $f(x, t) = 0$, then $f_y(t) = \int_0^\infty f(x, t)dx = 0$; else, if $t \in [0, e - 1]$, $f_y(t)$ is given as follow:

$$f_y(t) = \int_0^\infty f(x, t)dx = \int_0^\infty e^{-x(t+1)}dx = \frac{-1}{t+1} e^{-x(t+1)} \Big|_0^\infty = \frac{1}{t+1} \quad (1)$$

$$\text{So, } f_y(t) = \begin{cases} \frac{1}{t+1} & 0 \leq t \leq e - 1 \\ 0 & \text{otherwise} \end{cases}.$$

This indicates that the conditional probability density $f_{X|Y}(s|t)$ is only defined for $t \in [0, e - 1]$ (since it requires $f_y(t) \neq 0$ on the region). Which, one has the following:

$$t \in [0, e - 1], \quad f_{X|Y}(s|t) = \frac{f(s, t)}{f_y(t)} \begin{cases} (t+1)e^{-s(t+1)} & s \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Hence, given $Y = y$ (where $y \in [0, e - 1]$), the expectation of X is given as follow:

$$\mathbb{E}[X|Y = y] = \int_0^\infty x \cdot f_{X|Y}(x|y)dx = \int_0^\infty x(y+1)e^{-x(y+1)}dx = -xe^{-x(y+1)} \Big|_0^\infty + \int_0^\infty e^{-x(y+1)}dx \quad (3)$$

$$= -\frac{1}{y+1} e^{-x(y+1)} \Big|_0^\infty = \frac{1}{y+1} \quad (4)$$

□

Question 2. 8.3.18:

Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} n(n-1)(y-x)^{n-2} & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the conditional expectation of Y given that $X = x$.

Proof. First, we'll calculate the marginal density of X (where $X, Y \in [0, 1]$): Given $x < 0$ or $x > 1$, since $f(x, y) = 0$, then $f_X(t) = \int_0^1 f(t, y) dy = 0$. Else, if $0 \leq x \leq 1$, it's described as follow (Note: if $t > y$, then $f(t, y) = 0$):

$$f_X(t) = \int_0^1 f(t, y) dy = \int_t^1 n(n-1)(y-t)^{n-2} dy = n(y-t)^{n-1} \Big|_t^1 = n((1-t)^{n-1} - (-t)^{n-1}) \quad (5)$$

As a result, the conditional density $f_{Y|X}(s|t) = \frac{f(t,s)}{f_X(t)}$ is only defined on some subset of $t \in [0, 1]$. Which, fixing $X = x$ (or $t = x$), the conditional expectation of Y is given as follow:

$$\mathbb{E}[Y|X = x] = \int_0^1 y \cdot f_{Y|X}(y|x) dy = \int_x^1 y \cdot \frac{n(n-1)(y-x)^{n-2}}{n((1-x)^{n-1} - (-x)^{n-1})} dy \quad (6)$$

$$= y \cdot \frac{(y-x)^{n-1}}{((1-x)^{n-1} - (-x)^{n-1})} \Big|_x^1 - \int_x^1 \frac{(y-x)^{n-1}}{((1-x)^{n-1} - (-x)^{n-1})} dy \quad (7)$$

$$= \frac{(1-x)^{n-1}}{((1-x)^{n-1} - (-x)^{n-1})} - \frac{(y-x)^n}{n((1-x)^{n-1} - (-x)^{n-1})} \Big|_x^1 \quad (8)$$

$$= \frac{(1-x)^{n-1}}{((1-x)^{n-1} - (-x)^{n-1})} - \frac{(1-x)^n}{n((1-x)^{n-1} - (-x)^{n-1})} \quad (9)$$

□

3 ND

Question 3. 8.4.8:

Let X and Y be independent (strictly positive) gamma random variables with parameters (r_1, λ) and (r_2, λ) respectively. Define $U = X + Y$ and $V = X/(X + Y)$.

- (a) Find the joint probability density function of U and V .
- (b) Prove that U and V are independent.
- (c) Show that U is gamma and V is beta.

Proof.

□

4 ND

Question 4. 8.Rev.16:

A point is selected at random from the bounded region between the curves $y = x^2 - 1$ and $y = 1 - x^2$. Let X be the x -coordinate, and let Y be the y -coordinate of the point selected. Determine if X and Y are independent.

Proof.

□

5 ND

Question 5. 8.Rev.21:

Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 1 & |y| < x, \ 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Show that $\mathbb{E}[Y|X = x]$ is a linear function of x while $\mathbb{E}[X|Y = y]$ is not a linear function of y .

Proof.

□

6

Question 6.

Proof.



7

Question 7.

Proof.

□

8

Question 8.

Proof.



9

Question 9.

Proof.

□

10

Question 10.

Proof.

